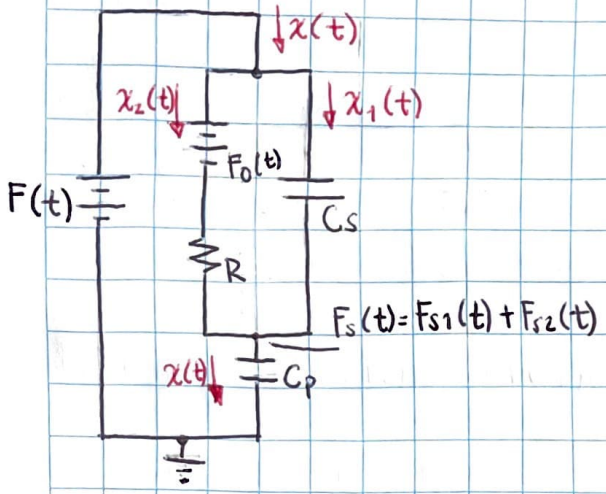
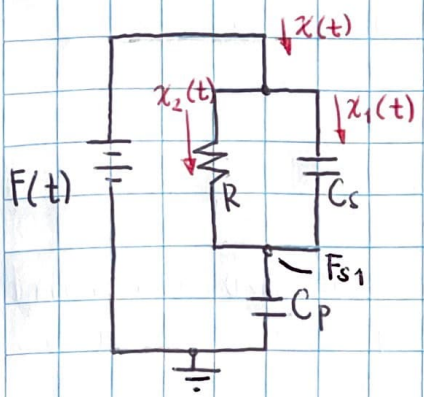
Práctica tres: Sistema musculoesquelético
Análisis por superposición

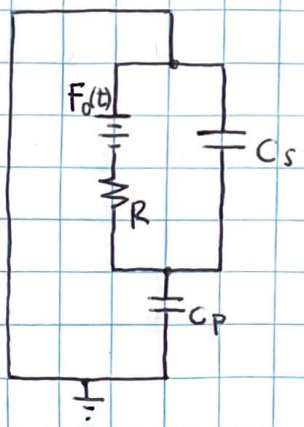
$x(t) = x_1(t) + x_2(t)$

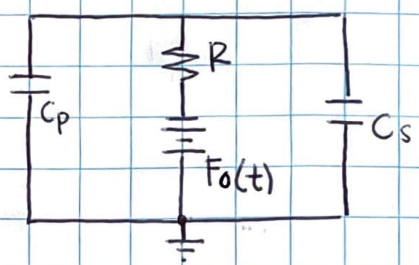


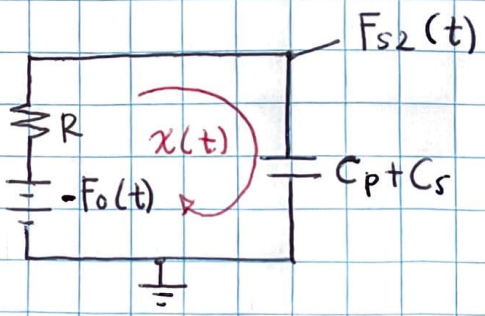
$F_0(t)$ en cortocircuito



F(t) en cortocircuito







Práctica 3
Análisis matemático

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición

Circuito 1. Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} - \frac{F(t) - F_{s1}(t)}{R}$$

Aplicando transformada de Laplace

$$C_p s F_{s1}(s) = C_s s[F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{S1}(s) = \frac{\frac{C_s R S + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{S1}(s) = \frac{C_s R S + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando $F(t)$

Recordando que $F_0(t) = \alpha F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{S2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

Transformada de Laplace

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{S2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) x(s) = -\frac{1}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s} x(s)$$